

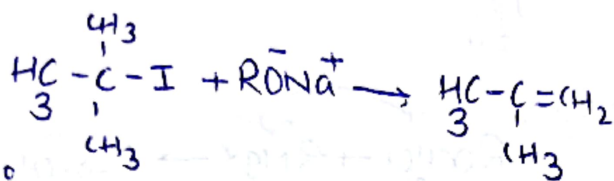
ETHERS

Preparations: $\underbrace{1, 2, 3}$



3
1° or 2° halides only

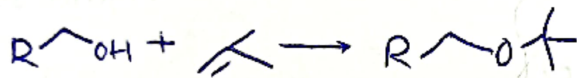
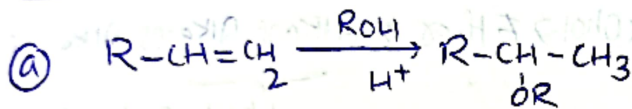
if it is



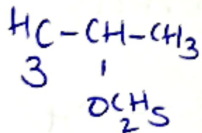
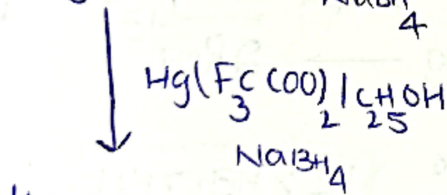
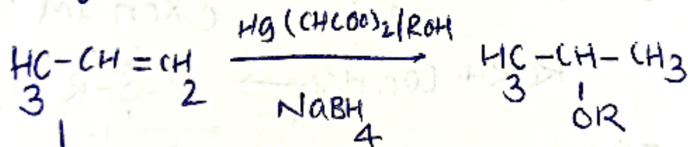
2. If it is 3° halides Elimination.

3° → E eliminations

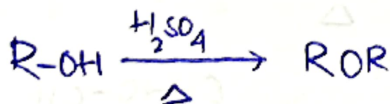
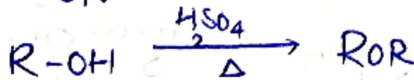
(ii) From alkenes:



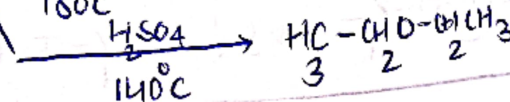
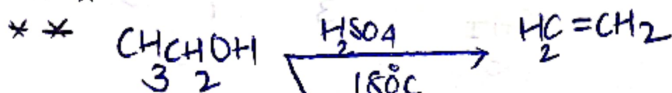
(b) alkoxy Mercuration:



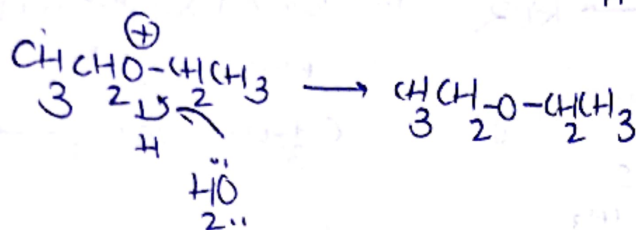
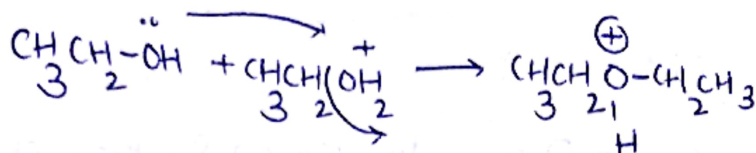
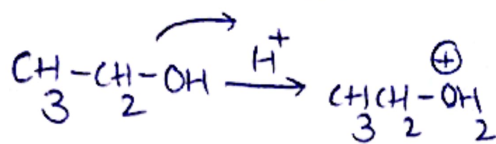
(iii) From alcohols : (H_2SO_4 is dehydrating agent
Removal of water)

 $i S_N^2$ 

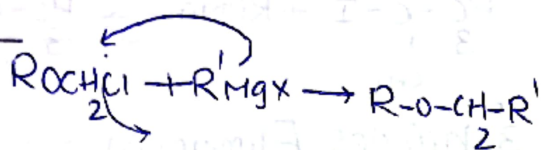
2° or 3° S.N.



Inter Molecular
dehydration.



4. RMgX:



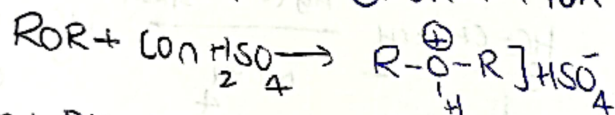
Alcohol : Hydrogen Bonding

Ether : Dipole Moment

Alcohol > Ether > Alkane, Alkenes, Alkynes
Hydrocarbons.

Properties:

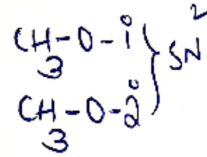
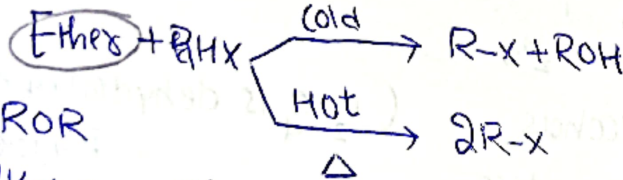
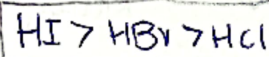
1. Ether + conc $\text{H}_2\text{SO}_4 \rightarrow$ Oxonium ion



2. Ether + Dil $\text{H}_2\text{SO}_4 \rightarrow 2\text{ROH}$

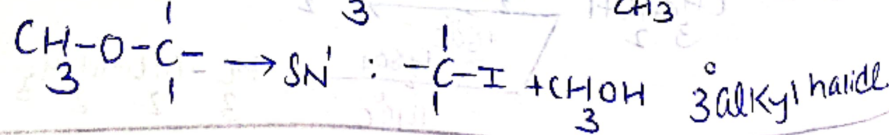
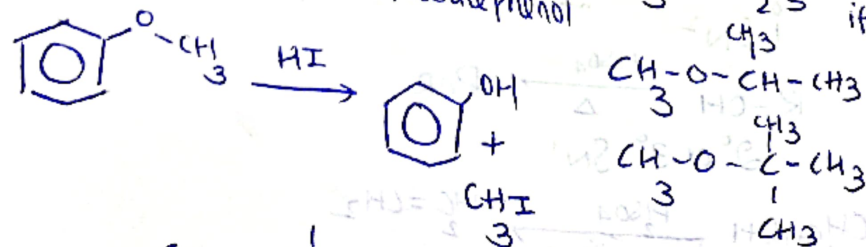
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3. Cleavage of Ether:

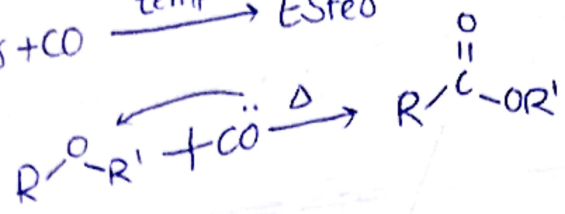


$\therefore$ alkyl alkyl ether on cleavage must produce phenol

One side $\rightarrow \text{CH}_3$
Other side $\rightarrow \text{CH}_3$ if it is alkyl



7. Ether + CO $\xrightarrow{\text{temp}}$ Ester

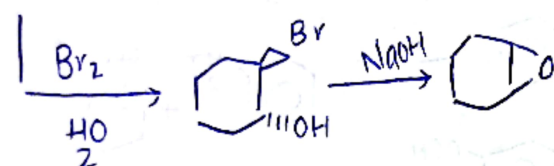
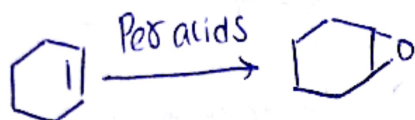


- A cyclic ether where ether forms three atom ring, Equilateral triangle structure

EPOXIDES

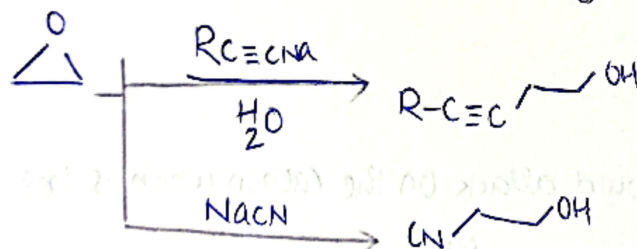
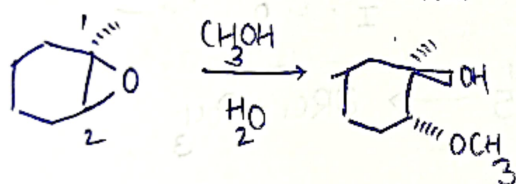
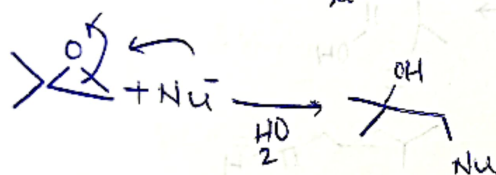


Epoxide ...



Ring Epoxide:

1. Nu attack: (Nu attacks at less hindered side)



2. Acid Catalysed Ring Opening:



Nu attacks from Stable C⁺

